First Topic:

IT Roles:

- **Developer**
- **Designer**
- **Researchers**
- **Systems Analyst**

What is System Analysis

- **This came from two greek words where they mean respectively “to take apart” and “to put together”**
- **Analysis is defined as the procedure by which we break down an intellectual or substantial whole into parts.**
- **Synthesis is defined as the procedure by which we combine separate elements or components in order to form a coherent whole.**
- **it is defined as explicit formal inquiry carried out to help a decision maker identify a better course of action and make a better decision than he might otherwise have made.**
- **System analysis is an explicit formal inquiry carried out to help a decision maker identify a better course of action and make a better decision than he might otherwise have made.**
- **Systems analysis is a problem-solving technique that decomposes a system into its component pieces for the purpose of studying how well those component parts work and interact to accomplish their purpose.**
- **This is a process used in the design of new systems. Systems analysis follows stages of investigation, design and implementation.**
- **Each stage should involve close consultation with potential users, in the various functional areas of the organisation, to ensure that their information and operational requirements are met.**

Where do we apply Systems Analysis?

What are the reasons to create system projects?

- **Improved service**
- **better performance**
- **More information**
- **Stronger controls**
- **Reduced Costs**

When to use SAD

- **to correct problem in existing system**
- **to improve existing system**
- **Usher in a new system**
- **Outside group may mandate change**
- **Competition can lead to change**

System Project overview:

- **Scope definition - is the project worth looking at**
- **Problem Analysis - *Is a new system worth building?***
- **Requirements Analysis - *What do the users need and want from the new system?***
- **Logical Design - *What must the new system do?***
- **Decision Analysis/Tradeoffs Analysis - what is the best solution?**

What to ask for on a client

- **ask for documentations or something about the client's processes**
- **Let the client articulate their processes**
- **do not adjourn the meeting until we are coherent bruh**

SDLC

- **The SDLC in system analysis and design aims to produce a high-quality system that meets or exceeds customer expectations, reaches completion within time and cost estimates, works effectively and efficiently in the current and planned Information Technology infrastructure, and is inexpensive to maintain and cost-effective to enhance.**

- **Phases of SDLC**
  - **Requirement Analysis**
  - **Defining**
  - **Designing**
  - **Coding**
  - **Testing**
  - **Maintenance**
- **Requirement Analysis**
  - **Review project requests**
  - **Allocate Resources**

    - **Identify project dev team**
    - **identifying business value**
    - **analyze feasibility**
    - **develop work plan**
    - **staff the project**
    - **control and direct project**
        - **Systems Development Life Cycle Feasibility Study**
            - **Operational Feaasibility**
            - **Technical Feasibility**
            - **Schedule(time) feasibility**
            - **Economic feasibility**
- **Defining**

    - **Data Gathering**
    - **Study Current System**
    - **Determine user requirements**
    - **recommend a solution**
    - **gathering business requirements**
    - **requirements definition**
    - **Conduct preliminary investigation.**
    - **Determine exact nature of problem or improvement and whether it is worth pursuing.**
    - **Findings are presented in feasibility report (feasibility study)**
    - **Perform detailed analysis activities:**
    - **Study current system**
    - **Determine user requirements**
    - **Recommend solution**
    - **Gathering business requirements**
    - **Requirements definition**
- **Root Cause Analysis**

    - **way of identifying underlying source of a process or product failure so that the right solution can be identified**
    - **Use 5 why's analysis**
- **Designing**

    - **Assesses feasibility of each alternative solution**
    - **Recommends the most feasible solution**
    - **How system will be developed**
    - **Design selection**
    - **Architecture Design**
    - **Interface Design**
    - **Data Storage Design**
    - **Program Design**

- **Construction/Coding**

    - **Program building Develop programs**
    - **Program and system testing**
- **Testing**

    - **Identify errors and enhancements**
    - **pre implementation system reviews**
    - **Conduct pre-implementation system reviews**
    - **Identify errors and enhancements**
- **Deployment**

    - **Installation**
    - **Implementation Strategy**
- **Maintenance**

    - **Monitor System Performance**
    - **Versioning**
- **SDLC Feasibility Study**

    - **Technical feasibility**
    - **Operational Feasibility**
    - **Schedule Feasibility**
    - **Economic Feasibility**

Business Case:

- **Structured justification for a proposed system**
- **Includes problem statement, objectives & stakeholders**
- **Cost/Benefit Analysis, Constraints, Risks, Solution Options**

Analyzing a business problem

- **Stakeholders: Internal and external actors**
- **Objectives: Tangible and intangible goals**
- **Triple Constraints: Cost, Scope and Time**

Techniques for Gathering Requirements

- **Document Analysis: Review existing documents and procedures**
- **Surveys: Data collection through questionnaires usually with close-ended questions.**
- **Shadowing: Direct observation**
- **Interviews: Data collection through one-on-one talk with stakeholders usually to verify collected data from other data gathering techniques.**
- **Workshops: Focus Group Discussions**
- **Prototyping: Initial proposed output**

Brainstorming with your group

- **A group problem-solving technique where participants discuss a problem together and suggest solutions or ideas, freely and without judgement**
- **Tips on Brainstorming:**
  - **Defer criticism**
  - **Encourage all ideas**
  - **Aim for quantity over quality**
  - **Build on others' ideas (piggybacking)**
  - **Record everything**
  - **Clarify the prompt clearly**
  - **Allow for silent thinking first**

Preliminary System Requirements

- **High-level descriptions of what the system must do**
- **Functional: System actions**
- **Example: System must allow digital document uploads**

Business Case Requirements

- **Requirement:**
  - **Root Cause Analysis**
  - **Project Title**
  - **Users**
  - **Objectives(General and Specific)**
  - **Preliminary System Requirements**

SPECIFIC, MEASURABLE, ATTAINABLE, REALISTIC, TIME-BOUND(SMART)